MCM Problem A: Paper Review & Rescue Guide

Team #2630023

Generated from Review Report

2026 年 2 月 2 日

1 Strengths and Weaknesses

Strengths

Our framework forms an explicit continuous-time closed loop from usage inputs to port power, then to ECM current/voltage, and finally to SOC and TTE events, satisfying the problem' s requirement for a non-black-box SOC(t) model (Eqs. (2), (9), (10), (33)). The first-order Thevenin ECM with an analytic $P \to I$ mapping and an event-based TTE definition captures realistic early shutdown via voltage cutoff or infeasible port power (Eqs. (3)–(5), (33)–(34)) while remaining computationally lightweight. Key submodules are validated using measured V–I discharge traces (Fig. 4), LOOCV for signal-strength cases (Fig. 9), NASA thermal data (Fig. 14), and a constant-current analytic special case (Fig. 18).

Weaknesses

Several power coefficients and behavior templates are device-/app-dependent; applying the model to a specific phone requires calibration from openly available measurements, and the manuscript should explicitly state the open-license provenance of all data used (TBD, per COMAP requirement). Except for the temperature dependence $R_0(T_b)$, ECM parameters are held constant and aging is modeled only as capacity fade; SOC-dependent parameters and impedance growth may affect accuracy near end-of-discharge (Sec. 2, Sec. 7.4). The lumped thermal RC model neglects spatial temperature gradients and heat from non-battery components, so predictions under localized heating or strong OS throttling may deviate (Sec. 5).

2 Conclusions and Recommendations

We developed a continuous-time smartphone battery drain model that couples SOC dynamics with a first-order Thevenin ECM, modular system power decomposition, and electro-thermal feedback, and computes TTE via event-based criteria (Secs. 3–7). Across scenarios, cold conditions and weak cellular signal produce the largest TTE losses (e.g., at 100% SOC under gaming, 10°C reduces TTE by 37% vs. 13% at 40°C; a −60 to −110 dBm signal drop can reduce gaming TTE by about 40%), while CPU-coefficient uncertainty and the polarization-branch switch have comparatively small effects (∼2–7% and ∼1%, respectively; Secs. 7.2, 8.1). **Recommendations (based on our results):** (i) reduce brightness / enable aggressive adaptive dimming; (ii) use signal-aware networking (prefer Wi-Fi and batch background transfers under poor RSSI); (iii) apply cold-aware usage and OS power management, because these levers align with the dominant sensitivity drivers identified by the model (Secs. 7.2, 8.2).

Errata and Corrections to the Previous Assistant Output (Accuracy Check)

- **Fixed a figure cross-reference error present in your paper:** in Sec. 4.2.2, the sentence "As illustrated in Figure 16" should read "As illustrated in Figure 7" (Fig. 16 is wake events; Fig. 7 is the display robustness plot).
- **Unified data-rate notation:** your Eq. (14) text says data transfer rate $D(t)$ but the formula uses $R_{\text{data}}(t)$; the minimal fix is to use $D(t)$ consistently in Eq. (14) and throughout B2L (Sec. 6).
- **Clarified Fig. 22 caption mismatch:** Fig. 22 is a radar-style attribution/driver visualization, but its caption duplicates Fig. 21; the caption should be updated to reflect driver ranking/attribution (and the incomplete sentence "26.4/" must be removed or completed).
- **Adjusted my Notation table to match the minimal renaming plan:** keep $D(t)$ as data rate, rename degradation state $D \to D_{\text{deg}}$, and keep $I(t)$ reserved for current (use $\mathbf{1}[\cdot]$ for indicators).
- **Removed time-estimate language from the Patch Plan:** I previously included minute-level time estimates; this revised version omits them and focuses on concrete paste-ready patches.

评审报告（约 300 字）

该稿合规性总体可判为 Pass：明确给出连续时间 SOC 微分方程与 Thevenin 等效电路（ECM），并以端口功率闭环求解 $I(t)$、用事件准则计算 TTE；同时提供了 V–I 对照、LOOCV、NASA 热数据与恒流特例验证，并完成参数与使用扰动敏感性。当前最大扣分点集中在“交卷前清理”：第 9/10 章空白（Strengths/Limitations/Recommendations/Conclusions 缺失），出现 [Citation]/[?] 与 “26.4/” 残句，符号多义（$D/I/T_b$）与图号引用错误（Fig. 16↔Fig. 7）、Fig. 22 caption 重复会触发严谨性警报。若不修，整体像 HM；按补丁补齐关键章节、清理占位符并统一符号与 caption 后，竞争力可明显上升。

修改前预计奖项与排名（1000 份作品）

Estimated award (before fixes): Honorable Mention (HM) is the most plausible outcome, mainly due to the blank Sec. 9–10 and the placeholder/cutoff text. **Estimated rank (before fixes):** roughly **350 ± 100** out of 1000 (very uncertain; could drift higher if a judge overlooks the blanks, or lower if the placeholders are treated as a credibility failure).

A. 总评（约 300 字中文总结）

你们的核心建模路线（连续时间 SOC ODE + Thevenin ECM + 功耗分解 + 热-电耦合 + 事件型 TTE）是合规且具备高分潜力的；验证链条也覆盖了 V–I 对照、LOOCV、NASA 热数据与恒流特例。但当前稿件存在几处 “肉眼扣分/可能直接丢分” 的硬伤：Sec. 9/10 为空、出现占位符引用（如 [Citation]/[?]）与残句 “26.4/”、符号多义冲突（$D/I/T_b$）以及多处 caption 不带结论。在 “不能大改” 的约束下，不需要推翻模型，只要补齐缺失章节、完成符号统一、修正明显图号/caption 错误并清理占位符，就能显著提升评分与可信度。

B. 题目要求映射表（表格）

Problem A requirement	Where in your paper	Status	Minimal patch
Explicit continuous-time SOC model (no black-box ML)	Sec. 3.1.1 Eqs. (1)–(2); Sec. 7.1 Eq. (32)	OK	Add one sentence in Summary/Intro emphasizing ODE + port-power closure.

ECM / physically interpretable battery model	Sec. 3.1.2–3.1.4 Eqs. (3)–(9); Fig. 3	OK	—
Power decomposition (screen/CPU/network/background; coupling allowed)	Sec. 4.2 Eqs. (11)–(15); Sec. 6 B2L templates	OK	Add one sentence clarifying GPS/Camera loads are represented via templates or can be added as explicit $P_{\mathrm{GPS}}(t)$ etc (TBD).
Temperature effects	Sec. 5 Eqs. (23)–(26); Figs. 13–14; Sec. 7.2 Figs. 19–20	OK	Fix the text "ambient temperature T_b" $\rightarrow T_{\mathrm{amb}}$ (Sec. 7.2).
TTE prediction across scenarios/initial SOC	Sec. 7.1–7.4 Eqs. (33)–(36); Figs. 19–21, 23–24	OK	—
Compare to observed/plausible behavior + uncertainty + discuss where good/poor	Fig. 4; Fig. 9; Fig. 14; Fig. 18; Sec. 8	Mostly OK	Add Limitations sentences explaining when error may grow (cold/low SOC/high load) and note uncertainty sources (Sec. 9).
Identify biggest/smallest drivers	Sec. 7.3 (ranking); Sec. 8 (sensitivity)	OK	Remove/complete "26.4/" line; align Fig. 22 caption with driver attribution.
Sensitivity to assumptions/parameters/usage fluctuations	Sec. 8.1–8.2 Eqs. (37)–(38); Figs. 25–26	OK	—
Recommendations (user + OS)	**Missing (Sec. 10 empty)**	**FATAL**	Add 3–6 recommendation sentences tied to Figs. 20–21, 25–26 (paste-ready text above).
Strengths / limitations / extensions	**Missing (Sec. 9 empty)**	**FATAL**	Add short strengths/weaknesses (paste-ready text above).

Open data with open license stated	NASA dataset mentioned; license not stated	Major gap	Add explicit statement: data are publicly available and license is open (TBD: verify license text).
Figures/tables have conclusion-style captions	Many captions are descriptive only	Major	Append "showing that ..." conclusion clause to each caption.
Notation/symbol consistency (no multi-meaning)	D vs. $D(t)$; $I(\cdot)$ vs. $I(t)$; T_b vs. T_{amb}	Major	Rename degradation $D \to D_{\text{deg}}$; indicator $I(\cdot) \to \mathbf{1}[\cdot]$; ambient T_{amb}.

C. 最高杠杆问题 Top-10（Fatal/Major/Minor）

Fatal

1. Sec. 9 is empty (strengths/limitations/extensions required).
2. Sec. 10 is empty (conclusions + recommendations required).
3. Placeholder/cutoff text exists: "[Citation]", "Ding et al. [?]", and "26.4/".

Major

1. Symbol conflicts: D (degradation) vs. $D(t)$ (data rate); $I(\cdot)$ indicator vs. $I(t)$ current; T_b used as ambient in Sec. 7.2 text.
2. Many captions are not conclusion-driven ("what" but not "so-what").
3. Thermal consistency: with $C_{\text{th}} = 72$ and $R_{\text{th}} = 3.0$, $\tau_{\text{th}} = 216$ s, but text states $\tau_{\text{th}} \approx 375$ s (Sec. 5.4).
4. Incorrect/ambiguous internal references: Sec. 4.2.2 cites Fig. 16 but should cite Fig. 7.

Minor

1. P_{base} vs. $Pbase$ inconsistent; V_{oc} vs. VOC inconsistent.
2. Fig. 22 caption duplicates Fig. 21 but figure content is attribution/driver radar.

D. 可直接替换的文本（英文）

D1) Summary Sheet（整页替换，英文）

Lithium-Ion Smartphone Battery Drain: A Continuous-Time, Interpretable Model
Team Control Number: 2630023

Model (one sentence). We model smartphone state-of-charge $z(t)$ in continuous time using a charge-balance ODE driven by a first-order Thevenin equivalent-circuit model (ECM) and a physics-based port-power balance $P_{\text{bat}}(t) = V(t)I(t) = P_{\text{sys}}(t) - P_{\text{chg}}(t)$, where P_{sys} is decomposed into display, CPU, network, and background components with explicit coupling.

Governing dynamics (compact). SOC: $\dot{z}(t) = -I(t)/(3600Q_{\text{nom}})$. ECM: $V(t) = V_{\text{oc}}(z) - I(t)R_0 - v_1(t)$, $\dot{v}_1(t) = -(1/\tau)v_1(t) + (R_1/\tau)I(t)$, $\tau = R_1C_1$. Power-to-current mapping: the ECM quadratic yields an analytic $I(t)$ as a function of (z, v_1, P_{bat}). TTE event: $\text{TTE} = \inf\{t : \ z(t) \le z_{\min} \text{ or } V(t) \le V_{\text{cut}} \text{ or } \Delta(t) < 0\}$, with discriminant $\Delta(t)$.

Validation (one sentence). We validate the ECM voltage/current response against measured discharge traces (Fig. 4), validate network-power parameters using LOOCV with most relative errors within 10% (Fig. 9), validate thermal dynamics against NASA discharge temperature data (Fig. 14), and confirm SOC/TTE consistency using a constant-current analytic special case (Fig. 18).

Key quantitative findings (model-based). (1) Under gaming at 100% SOC, 10°C reduces TTE by 37% while 40°C reduces TTE by 13% relative to 25°C (Sec. 7.2, Fig. 20). (2) A signal degradation from −60 dBm to −110 dBm can reduce gaming TTE by ∼40% (Sec. 7.2, Fig. 21). (3) Sensitivity ranking: α_{signal} dominates TTE sensitivity (52.45%–56.78%), followed by display nonlinearity γ (20.31%–27.43%) and α_{data} (15.80%–20.85%), while α_{cpu} contributes only 2.04%–6.63%; neglecting polarization ($v_1 \equiv 0$) changes TTE by about 1% (Sec. 8.1). (4) Brightness leverage is strongest: $0.8B(t)$ changes TTE by 30.43%–47.52% (Sec. 8.2). (5) Signal improvements matter more at low SOC: $S(t) + 10$ dBm yields an 11.30%→3.15% gain as SOC increases, while $S(t) - 10$ dBm causes ∼7% loss with little SOC dependence (Sec. 8.2).

Actionable recommendations (user + OS). (Recommendation, based on findings) Reduce brightness / enable aggressive adaptive dimming. **(Recommendation)**

Prefer Wi-Fi and avoid high-throughput tasks under weak RSSI; batch/delay background transfers when signal is poor. **(Recommendation)** Apply cold-aware usage and OS policies because cold penalties dominate heat penalties in TTE.

Keywords. continuous-time SOC; equivalent circuit model (ECM); power decomposition; signal strength; thermal–electrical coupling; time-to-empty (TTE)

D2) Introduction（替换 Sec. 1.1–1.3，英文）

1.1 Problem Background

Smartphone battery life can vary widely because discharge couples time-varying device power demand (brightness, CPU workload, network activity, and background services) with battery electrochemical limits (voltage drop, polarization, and temperature-dependent resistance). A predictive model should therefore represent battery state evolution continuously in time, map realistic usage patterns into power demand, and capture physical mechanisms that trigger early shutdown such as voltage collapse at low SOC or under cold conditions.

1.2 Restatement of the Problem

We aim to: (i) build an explicit continuous-time model for SOC $z(t)$ under realistic usage; (ii) decompose $P_{\text{sys}}(t)$ into major subsystems and allow coupling; (iii) compute TTE under varying initial SOC and scenarios and compare to observed or plausible behavior with uncertainty discussion; (iv) identify dominant drivers and which factors change TTE the most/least; (v) analyze sensitivity to assumptions, parameters, temperature, and capacity fade; and (vi) translate insights into practical recommendations for users and OS power management.

1.3 Our Approach and Contributions

We employ a first-order Thevenin ECM and a physics-based port-power balance to map usage-dependent system power $P_{\text{sys}}(t)$ and charging inflow $P_{\text{chg}}(t)$ to current/voltage and SOC in continuous time. TTE is defined using event-based criteria (SOC threshold, voltage cutoff, and feasibility of the port-power relation) to capture realistic shutdown behavior. Our contributions are: (1) an explicit SOC–ECM–power closed loop (Secs. 3–4, 7); (2) an interpretable modular power decomposition plus a behavior-to-load interface for scenarios (Secs. 4 and 6); and (3) multi-level validation and plausibility checks (Figs. 4, 9, 14, 18).

D3) Assumptions（替换 Sec. 2，英文）

Unless otherwise stated, we use baseline conditions $T_b = 25°\text{C}$ (298 K), a fresh battery ($D_{\text{deg}} = 0$), and no self-discharge ($k_{\text{sd}} = 0$). (1) The battery is represented by SOC $z(t) \in [0, 1]$ governed by charge conservation (Eq. (2)). (2) A first-order Thevenin ECM is sufficient for smartphone time scales (Eqs. (3)–(5)). (3) Under baseline conditions, ECM parameters are constant; the structure accommodates $R_0(T_b, D_{\text{deg}})$ etc. for extensions (Sec. 2, Sec. 5, Sec. 7.4). (4) Net battery power satisfies $P_{\text{bat}}(t) = P_{\text{sys}}(t) - P_{\text{chg}}(t)$ and $P_{\text{bat}}(t) = V(t)I(t)$ (Eqs. (6), (10)). (5) System power decomposes into major subsystems (Eq. (11)); additional terms (GPS/camera) can be added modularly (TBD if needed). (6) Battery temperature follows a lumped thermal RC model (Eq. (23)); temperature enters primarily through $R_0(T_b)$ (Eq. (26)). (7) Aging is modeled as capacity fade only (Sec. 7.4). (8) TTE is an event-based failure time using SOC depletion, voltage cutoff, and feasibility discriminant (Eq. (33)).

D4) Notation Table（英文，LaTeX longtable）

Symbol	**Meaning (units)**
t	time (s)
$z(t)$	state of charge, SOC (–)
z_0	initial SOC (–)
$z_{\min}$	SOC threshold for shutdown event (–)
Q_{nom}	nominal battery capacity (Ah)
$Q_{\text{nom},0}$	beginning-of-life nominal capacity (Ah)
s	capacity-fade factor $Q_{\text{nom}}/Q_{\text{nom},0}$ (–)
D_{deg}	degradation state variable (–)
$I(t)$	battery current (A), positive for discharge
$V(t)$	terminal voltage (V)
$V_{\text{oc}}(z)$	open-circuit voltage (V)
R_0	ohmic internal resistance (Ω)
R_1	polarization resistance (Ω)
C_1	polarization capacitance (F)
$v_1(t)$	polarization-branch voltage (V)
τ	polarization time constant R_1C_1 (s)
$P_{\text{sys}}(t)$	total system power (W)
$P_{\text{chg}}(t)$	external charging inflow at device-side reference plane (W)
$P_{\text{bat}}(t)$	net battery power $P_{\text{sys}} - P_{\text{chg}}$ (W)

P_{base}	baseline system power (W)
$P_{\text{disp}}(t)$	display power (W)
A_{screen}	screen emissive area (cm^2)
$B(t)$	normalized brightness (–)
α_{disp}	display power density coefficient (W/cm^2)
γ	brightness nonlinearity exponent (–)
$P_{\text{proc}}(t)$	processor power (W)
$P_{\text{proc,idle}}$	processor idle power (W)
$L_{\text{fg}}(t), L_{\text{bg}}(t), L_{\text{net}}(t)$	foreground/background/network CPU load indicators (–)
α_{cpu}	CPU load-to-power coefficient (W per unit load)
β_{net}	CPU–network coupling coefficient (–)
$P_{\text{net}}(t)$	network power (W)
$P_{\text{net,idle}}$	network idle power (W)
$D(t)$	data transfer rate / throughput (Mbps)
$S(t)$	signal strength (dBm)
α_{data}	data-rate coefficient (W/Mbps)
α_{signal}	signal-related power scale (W)
β_{signal}	signal sensitivity (dBm^{-1})
$P_{\text{bg}}(t)$	background/wakeup power (W)
$N_{\text{wake}}(t)$	wake-up event rate (events/s)
α_{wake}	energy per wake event (J/event)
N_{base}	baseline wake rate (events/s)
σ_{suspend}	screen-off background suppression factor (–)
$\text{plug_in}(t)$	charger connection indicator $\in \{0, 1\}$ (–)
$P_{\text{chg,wall}}(t)$	wall-side charger power tier (W)
η^*_{chg}	adapter/device efficiency (–)
$P^{\max}_{\text{adp}}(t)$	max device-side adapter power (W)
$P^{\max}_{\text{abs}}(t)$	max battery absorption power (W)
P^*_0	charging power constant (W)
$s_T(T_b)$	thermal throttling function for charging (–)
$s_z(z)$	SOC tapering function (–)
z_{knee}	SOC knee initiating taper (–)
σ_z	SOC taper decay parameter (–)
$T_b(t)$	battery temperature state (°C)
T_{amb}	ambient temperature (°C)
C_{th}	thermal capacitance (J/°C)
R_{th}	thermal resistance (°C/W)

τ_{th}	thermal time constant $R_{\text{th}}C_{\text{th}}$ (s)
V_{cut}	voltage cutoff threshold (V)
$\Delta(t)$	feasibility discriminant in quadratic current relation (V^2)
TTE	time-to-empty (s)
k_{sd}	self-discharge coefficient (TBD units; baseline set to 0)

E. 符号审计报告（冲突 → 修复 → Find/Replace）

E1) Conflicts found

- D is overloaded: $D = 0$ in Sec. 2 means "fresh battery", but $D(t)$ in Sec. 4.2.4 and Sec. 6 is data rate.
- I is overloaded: Eq. (21) uses $I(\cdot)$ as an indicator while $I(t)$ is battery current everywhere else.
- T_b is misused in text: Sec. 7.2 says "ambient temperature T_b", but ambient is T_{amb}; $T_b(t)$ is battery temperature state.
- γ_{net} appears once but the model defines β_{net} (Sec. 4.2.3 vs. text under Fig. 8).

E2) Minimal rename plan (lowest disruption)

- Rename degradation state $D \to D_{\text{deg}}$ (few occurrences).
- Replace indicator $I(\cdot)$ with $\mathbf{1}[\cdot]$ (matches your B2L algorithm style).
- Replace "ambient temperature T_b" with T_{amb} in Sec. 7.2 text.
- Replace $\gamma_{\text{net}} \approx 1$ with $\beta_{\text{net}} \approx 1$ (one sentence).
- Unify: P_{base} (not $Pbase$) and V_{oc} (not VOC).

E3) Find/Replace checklist

- "$D = 0$ (fresh battery)" → "$D_{\text{deg}} = 0$ (fresh battery)"
- $R_0(T_b, D) \to R_0(T_b, D_{\text{deg}})$ (and similarly R_1, C_1 if used)
- "indicator functions $I(\cdot)$" → "indicator functions $\mathbf{1}[\cdot]$"
- Sec. 7.2 text: "ambient temperature T_b" → "ambient temperature T_{amb}"
- "γ_{net}" → "β_{net}"

- "$Pbase$" → "P_{base}"; "VOC" → "V_{oc}"
- Eq. (14): $R_{\text{data}}(t) \to D(t)$ (consistent with the surrounding text and B2L)

F. Patch Plan（最高杠杆、最小改动；不含时间估计）

1. **Fill Sec. 9 and Sec. 10 (FATAL):** paste the three-sentence Strengths/Weaknesses and Conclusions+Recommendations text provided at the top of this file.
2. **Remove placeholders/cutoff text (FATAL):** replace "[Citation]" with "(TBD: add open-source citation/datasheet and license statement)", replace "Ding et al. [?]" with "(TBD: verify correct reference number among [7–8])", and delete/complete "26.4/".
3. **Fix thermal consistency (MAJOR):** in Sec. 5.4, set $\tau_{\text{th}} = R_{\text{th}}C_{\text{th}} = 216$ s and update the stability margin accordingly.
4. **Fix internal references/captions (MAJOR):** Sec. 4.2.2 "Figure 16" → "Figure 7"; Fig. 22 caption should describe driver attribution (not duplicate Fig. 21).
5. **Symbol unification (MAJOR):** apply the Find/Replace list in E3 and insert the Notation table.
6. **Caption "so-what" sweep (MAJOR):** append one conclusion clause ("showing that ...") to each figure/table caption to make every visual self-interpreting.
7. **Open-license statement (MAJOR, COMAP requirement):** add one sentence in Data/Validation sections: "All datasets used are publicly available and free to use under an open license (TBD: insert license text/URL)."

G. 最终清单打勾结果（仅含需要检查的项）

Model-layer checklist

- ✓ **Explicit continuous-time SOC model:** yes (Eqs. (1)–(2)).
- ✓ **At least one validation:** yes (Figs. 4, 9, 14, 18).
- ✓ **Sensitivity (1–2 key parameters):** yes (Sec. 8; Figs. 25–26).
- × **Strengths/limitations/extensions present:** no (Sec. 9 empty) → fixed by paste-ready Sec. 9 text.
- × **Recommendations present:** no (Sec. 10 empty) → fixed by paste-ready Sec. 10 text.

Writing-layer checklist (symbol consistency only)

- × Same symbol used for multiple meanings (D, I, T_b) → fixed by E3 Find/Replace.
- × Same meaning written with multiple symbols ($P_{\text{base}}/Pbase$, V_{oc}/VOC) → fixed by E3 Find/Replace.
- × Units explicitly stated and consistent (especially temperature) → fixed by Notation table + Assumptions baseline.